

		(change) in fluid surrounding body.... 1 each, max 2	B2	[2]
3	(a)	force x perpendicular distance	M1	
		(of the force) from the pivot.....	A1	[2]
	(b)	no resultant force (in any direction)	B1	
		no resultant moment (about any point).....	B1	[2]
	(c)	(i) correct direction in both.....	B1	[1]
		(ii)1 moment = $150 \times 0.3 = 45 \text{ N m}$ (1 sig. fig. -1)	A1	
		(ii)2 torque = 45 N m i.e. same is (i)	A1	
		(ii)3 $45 = 0.12 \times T$	C1	
		$T = 375 \text{ N}$	A1	[4]
4	(a)	(i)1 amplitude = $0.4(0) \text{ mm}$	A1	